PALEOMAGNETIC CHRONOLOGY, FLUVIAL PROCESSES, AND TECTONIC IMPLICATIONS OF THE SIWALIK DEPOSITS NEAR CHINJI VILLAGE, PAKISTAN[1]

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ABSTRACT

A 2800-m section of Siwalik strata containing the stratotypes for both the Chinji and Nagri formations has been dated by magnetic polarity stratigraphy, and the observed polarity zonation has securely been correlated with the Chron 17-7 segment of the time scale. The base of the section is the base of the Kamlial formation, which occurs near the top of Chron 17 (18.3 m.y.). The Kamlial-Chinji formation boundary occurs in the middle of Chron 15 (14.3 m.y.), the Chinji-Nagri boundary near the bottom of Chron 10 (10.8 m.y.), and the Nagri-Dhok Pathan boundary at the Chron 8–9 boundary (8.5 m.y.). The Siwalik deposits near Chinji Village consist of four distinct classes of fluvial cycles, each with a characteristic periodicity: a first order of 10^7 years, a second order of 10^6 years, a third order of 10^4–10^5 years, and a fourth order of 10^0–10^4 years. The river system responsible for Siwalik sedimentation flowed from west to east and had properties much like the modern Ganges River system. Sedimentation rates in the Chinji Village area increased gradually through time, going from 0.12 mm/yr in the Lower Siwaliks to 0.30 mm/yr in the Middle Siwaliks. At about 11 m.y. blue-green hornblendes suddenly appear in abundance in channel sands. The appearance of these hornblendes, accompanied with accelerated sedimentation, is the effect of uplift in the source area, specifically the Nanga Parbat region. The modern Chinji Village area is no longer a depositional site, but instead is involved in overthrusting, uplift, and erosion. As the Indian Plate has drifted north during the past 18 m.y., the Chinji Village area has been gradually transformed from a subdued karst topography into a major depositional center for Himalayan sediments and finally back into a sediment source area to complete a sedimentary cycle that has spanned some 20 m.y.

INTRODUCTION

The Siwalik sediments exposed near Chinji Village, Potwar Plateau area, Pakistan, include the stratotypes for the Nagri and Chinji Formations and the basal unit of the Siwalik sequence, the Kamlial Formation. The stratigraphic section studied and reported on here consists of two separate segments, the lower Chita Parwala section and the upper Gabhir Kas section (fig. 1). Although these segments lie some 6 km apart, they share two sandstone units which are physically continuous from one section to the other. Preliminary paleomagnetic data did not allow for a clear correlation between the magnetic stratigraphy of the section and the magnetic time scale (N. Johnson et al. 1982).

Extensive and improved resampling has resulted in 93 new paleomagnetic sites. Together with the 66 sites previously reported, these sites form a data base which now allows us to make age assignments based on the magnetic polarity time scale and to assess the sedimentary and tectonic implications of this chronology.

SIWALIK STRATIGRAPHY

Uplift and erosion of the Himalayas have produced an apron of fluvial sediments disposed along the southern flank of the mountains. In Pakistan and northwest India, the Miocene-Pleistocene equivalent of these sediments is generally called the Siwalik Group (fig. 1). In Pakistan the basal sand unit of the Siwalik Group is variously called the Kamlial or Murree Formation. A silty redbed unit usually overlies this basal sand and is designated the Chinji Formation. Together the Kamlial and Chinji Formations are called the Lower Siwaliks (Fatmai 1973). Their thickness ranges from several hundred to over a thousand meters. The Middle Siwaliks are

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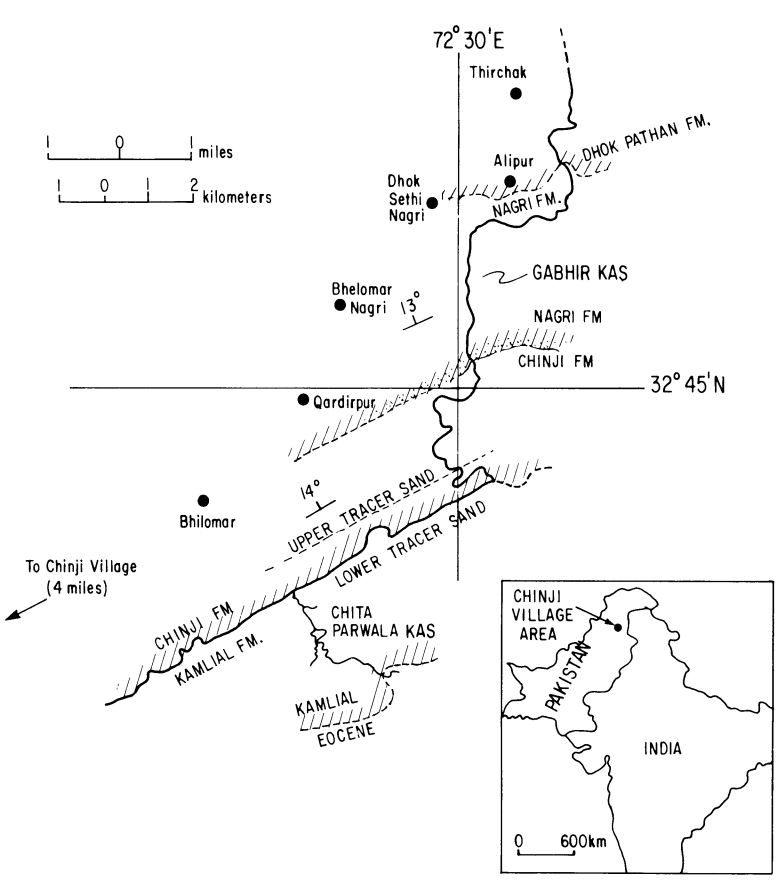


FIG. 1.—Location and geology of the study area. Stratigraphic section was measured normal to strike along river cuts, Chita Parwela Kas for the lower part of the section, and Gabhir Kas for the upper part. Stratigraphic units are continuously exposed between the Chita Parwela and Gabhir Kas sections.

defined as the combined Nagri and Dhok Pathan formations, which conformably overlie the Lower Siwaliks (Fatmai 1973). The Nagri Formation represents a return to predominantly sand deposition much like the Kamlial Formation. The Nagri sands are in turn replaced gradually by the brown silts of the overlying Dhok Pathan Formation. Over the Potwar Plateau area the Middle Siwalik strata generally exceed 800 m in thickness. It is noteworthy that the Lower and Middle Siwaliks are analogous fining-upward sequences, one following the other on a time scale of 10^6 years. A much smaller-scale feature of Siwalik sedimentation is the cyclic deposition of graded beds, collectively known as fluvial cycles. The frequency and interpretation of these larger and smaller cycles will be addressed later in this paper.

In previous work the Middle and Upper Siwalik units have been assigned absolute ages by means of magnetic polarity stratigraphy and isotopic dating (N. Johnson et al. 1982; G. Johnson et al. 1982; Opdyke et al. 1982; Tauxe and Opdyke 1982). A conspicuous lack of dating has characterized the Lower Siwaliks, especially the Kamlial Formation.

Over most of the Potwar Plateau the Kamlial Formation disconformably overlies marine limestone of Eocene age. This discomformity is well-exposed on the north flank of the Salt Range in the study area near Chinji Village. Here the Kamlial Formation is

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a 560 m-thick sequence of alternating sandstones and mudstones, with an average sandstone/mudstone ratio of 1.53. Many of the sandstones are multistoried and represent several successive depositional events. The lateral extent of the sandstone bodies is variable, with rapid facies changes occurring over relatively short distances. Two types of sandstones are represented in the Kamlial rocks: the first is massively bedded and well indurated, and the second is friable and poorly indurated with spheroidal weathering (Gill 1951). There do not appear to be distinct color differences between the two types of sandstone; the colors observed are generally greenish-grey.

The boundary between the Kamlial and Chinji Formations is marked by a rather sudden, qualitative shift in lithology which is easily mapped in the study area. Named by Pilgrim (1913), the Chinji Formation is the most recognizable unit of the Siwalik Group because of its preponderance of brightly colored silts and its sunken topographic expression relative to the Kamlial and Nagri sandstone cuestas which bound it. (Ironically, Chinji Village is situated on a cuesta of Nagri sandstone.) In our measured section the Chinji is 400–500 m thick, but farther west in the Chinji Village area it thickens to 700 m (Raza 1983). The Chinji Formation consists of greenish-gray sandstones and brownish-red mudstones; the sandstone/mudstone ratio averages 0.25, in striking contrast to the 1.53 sandstone/mudstone ratio in the Kamlial Formation. The basal contact of the sandstone units in the Chinji is commonly erosional and scoured into an underlying mudstone. The sand usually grades upward into silts and muds, which in turn are succeeded by an erosional contact with another sand unit. Recurring fluvial events such as these are designated "fluvial cycles" by Allen (1965). The thickness of these fluvial cycles is variable, ranging from 12 to 50 m. An extensive description of the Chinji Formation in the Chinji Village area is given by Raza (1983).

The Nagri Formation, named after the village complex of Sethi Nagri (fig. 1), conformably overlies the Chinji. In the study area the Nagri Formation is overwhelmingly composed of multistoried, green-gray sands whose total thickness is 700 m. Locally the

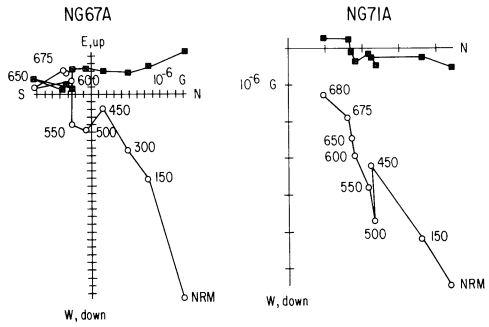


FIG. 2.—Zijderveld plots for normally magnetized (NG71A) and reversely magnetized samples (NG67A) during thermal demagnetization. Declination represented by filled square symbol, inclination by open circle symbol.

Nagri Formation also contains a bentonized air-fall tuff. Zircons from this tuff have produced a fission track date of 9.5 ± 0.5 m.y. (G. Johnson et al. 1982). In our measured section only the lowermost part of the Dhok Pathan Formation is exposed (fig. 1).

PALEOMAGNETIC METHODS

Between the base of the Kamlial and the top of the section, 159 paleomagnetic sites were sampled. Three block samples were collected from each paleomagnetic site using the orientation procedure of Johnson et al. (1975). The magnetic mineralogy from site to site is quite variable, but overall the principal magnetic carrier is detrital, specular hematite, much like that previously reported for the Siwalik sediments from other areas (Tauxe et al. 1980).

The thermal demagnetization curves of various rock types we deemed typical for our stratigraphic section are shown in figure 2. Specimen NG71A displays the three-component magnetization common to the Siwalik red beds (Tauxe and Badgley 1984). This behavior manifests itself as a decay of the remanence from NRM to 450°C, followed by an increase in remanent intensity to 500°C (an indication of the removal of a component in the opposite direction). Smooth decay to the origin from 500°C to 680°C indicates that the characteristic magnetization is carried by hematite. The behavior of a typical reversed specimen (NG67A) is shown in figure 2. Within this sample removal of a normal component, carried by fine-grained hematite, is

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completed by 600°C. Although the subsequent decay is not linear, the polarity direction of this sample is clearly reversed.

On the basis of pilot demagnetization curves such as these, we selected 600°C as a blanket treatment. We have followed the polarity classification scheme of Tauxe (1979). Paleomagnetic sites, consisting of three or more samples, were considered statistically significant if after thermal demagnetization at 600°C the Watson criterion was satisfied (Watson 1956). Such sites are called class A. Sites that could not be tested statistically, but whose polarity direction was not in doubt, were designated class B. Class B sites typically consisted of two samples with subparallel polarity directions, but a third sample that had a widely divergent or an ambiguous direction. Class B sites then are based on only two samples and as such are not amenable to the standard statistical tests. Of the 159 paleomagnetic sites collected, 99 satisfied the class A standard, 37 sites were classed B, and 23 sites were not used in this study because their polarity could not be determined with any certainty.

Figure 3 is a stereo plot of our class A data. Note that the means of the normal and reversed sites are antipodal, indicating that our 600°C treatment has successfully isolated a primary detrital remanence. We feel confident, therefore, that the rocks are faithfully recording the polarity of the ambient magnetic field at the time of sedimentation. The counterclockwise rotation of the data with respect to N-S is typical for Siwalik rocks of the Potwar area (Opdyke et al. 1982).

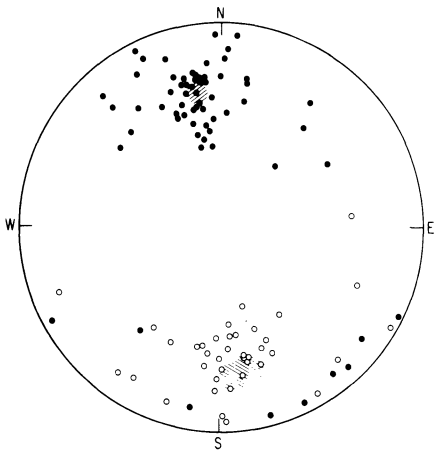


FIG. 3.—Stereographic projection of class A site vectors after thermal demagnetization. Filled symbols represent the lower hemisphere, open symbols the upper. Cones of 95% confidence (α_{95}) depicted by shaded pattern. Mean declination and inclination for all normally magnetized sites are 349.7° and +32.1°, respectively, k = 15.04, α_{95} = 5.07°, N = 56; for the reversely magnetized sites mean declination is 171.3°, mean inclination is −25.0°, k = 5.8, α_{95} = 9.3°, N = 48.

MAGNETIC POLARITY STRATIGRAPHY

In Figure 4 are the superposition of our paleomagnetic sites, their magnetic polarity, and the polarity transitions. With few exceptions, each polarity zone in our magnetic stratigraphy is defined by two or more paleomagnetic sites.

Paleomagnetic data from Gabhir Kas show 24 magnetic reversals which define 13 normal polarity zones and 12 reversed zones. In the Chita Parwala segment of the section there are 14 magnetic reversals, which define seven normal polarity zones and eight reversed zones. Significantly, the upper of our tracer sandstones is contained within one rather narrow normal polarity zone (N_7) in both sections. The sharing of N_7 in both the Chita Parwala section and the Gabhir Kas section provides a firm correlation between the two sections. Furthermore, the lower of the tracer sand units lies immediately above a very short reversed polarity interval, between Nq–N_8 (fig. 4). This short reversed polarity zone has been found in seven different stratigraphic sections between Gabhir Kas and Chita Parwala, so its continuity has been substantiated (Brannon 1983). Accordingly our stratigraphic tracers are time-parallel and provide reliable correlation between the Chita Parwala section and the Gabhir Kas section.

Perhaps the most striking feature of our magnetic polarity column (fig. 4) is the long normal polarity interval, N_0, associated with the Nagri Formation. It is noteworthy that in five other localities in the Potwar Plateau area, the Nagri is characterized by unbroken, normal polarity (N. Johnson et al. 1982). So, even though our magnetic sampling network appears relatively thin for the Nagri Formation, it does not seem likely that we missed any major polarity zones. The 13 consecutive normally magnetized sites which delimit N_0 provide a high level of sampling redundancy

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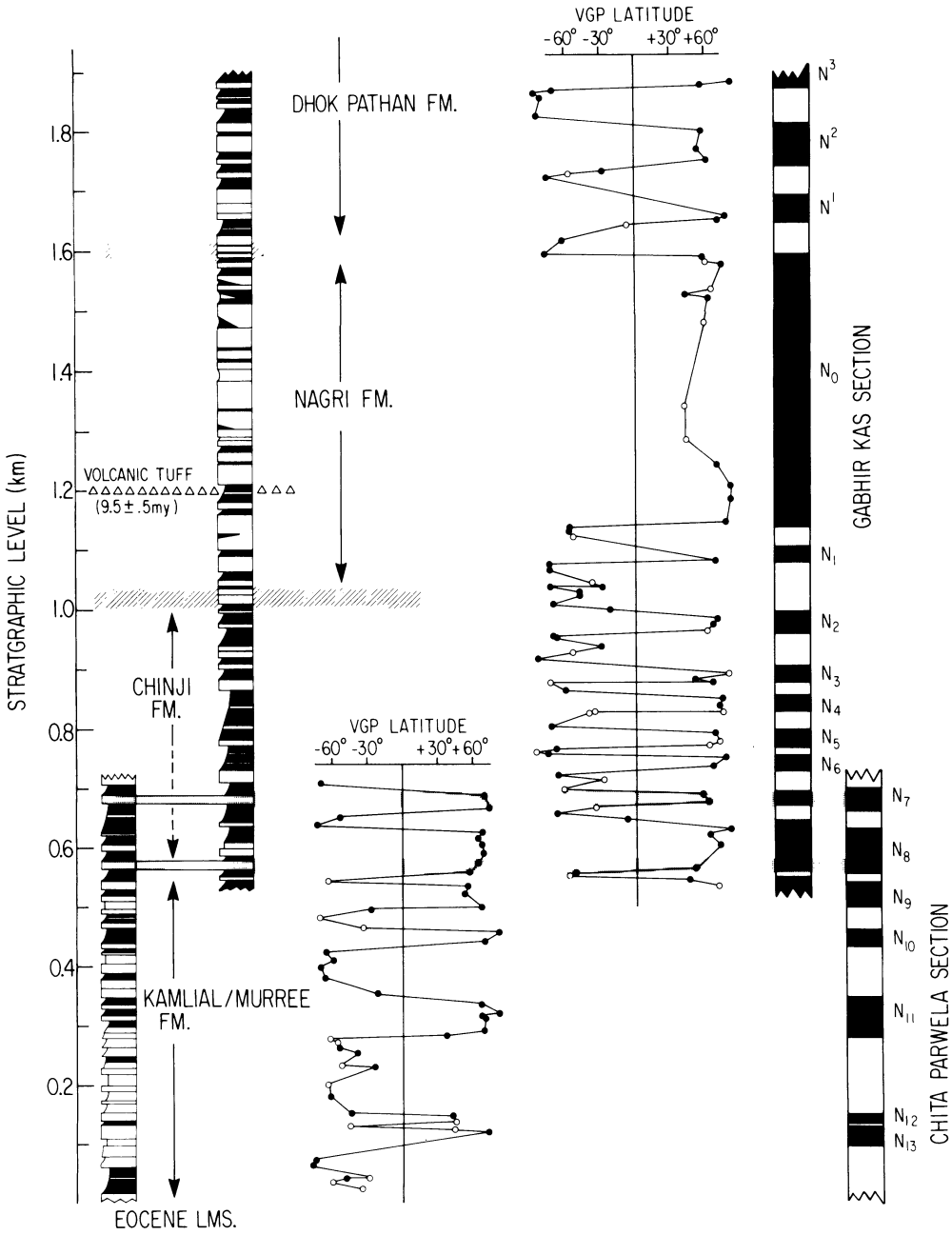


FIG. 4.—Superposition and magnetic polarity of sites in Chita Parwela section and Gabhir Kas section. Class A sites depicted as closed circles, class B as open circles. Tracer units between Chita Parwela and Gabhir Kas sections highlighted with shaded pattern.

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and strongly mitigate against N_0 being just a random extreme in our sampling program.

In contrast to the pervasive normal polarity of the Nagri Formation the Kamlial is predominantly of reversed polarity, while the Chinji Formation is distinguished by its decidedly complex magnetic polarity sequence (fig. 4).

In figure 5 is our correlation of the local magnetic polarity stratigraphy with the Mankinen and Dalrymple (1979) time scale. This correlation is based on several lines of evidence. First, a fission-track date of 9.5 ±0.5 m.y. from the Nagri Formation (G. Johnson et al. 1982) firmly identifies the long normal polarity in the Nagri Formation as Chron 9.

Second, it is possible to estimate statistically the time interval for our section (table 1). If the statistical behavior of the Earth's magnetic field is known, a statistical model allows us to calculate the amount of time represented in a given stratigraphic column from the number of paleomagnetic sites placed and the number of reversals found (Johnson and McGee 1983) based on a Poisson distribution of reversals through time. We know that the Earth's magnetic field over the past 10.6 m.y. demonstrably has reversed itself in a Poisson fashion (Cox 1969). Prior to 10.6 m.y., however, the data for the magnetic time scale are less complete and their statistical properties less certain. Nevertheless, a recent study by Lowrie and Kent (1983) has shown that when all pertinent data are considered, the distribution of reversals over the past 40 m.y. is not statistically different from a Poisson distribution. Therefore, for the purposes of our study we will assume that a Poisson distribution of reversals was in force while our stratigraphic section was being deposited.

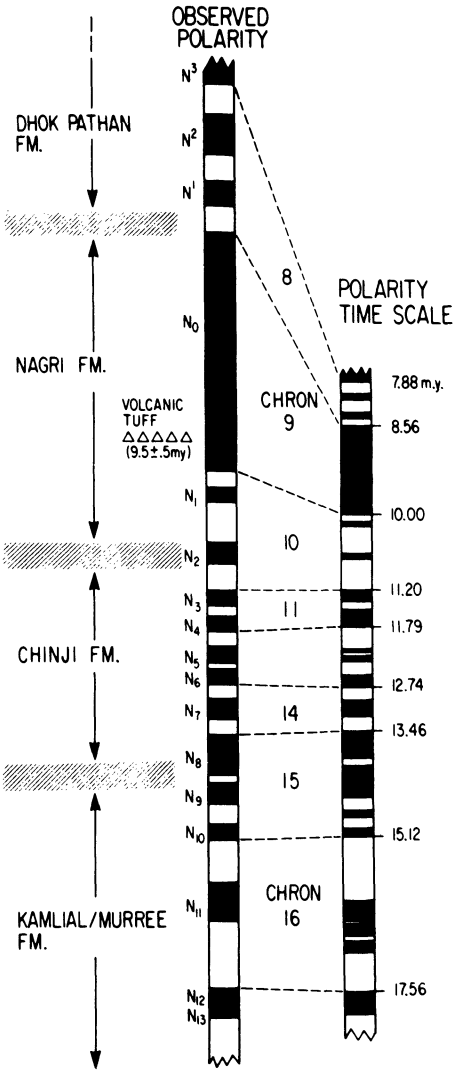


FIG. 5.—Correlation of the observed magnetic polarity zonation (fig. 4) with the magnetic reversal time scale of Mankinen and Dalrymple (1979).

Allowing for the duplication of samples between our two stratigraphic sections, our sampling array of 92 sites uncovered 27 reversals below N_0 (fig. 4). From the sampling model (table 1) we thus expect that 8.3 ±1.3 m.y. (1σ) of time are spanned by the strata below N_0. As discussed above, N_0 is correlated with Chron 9, whose base is 10.00 m.y. in age (Mankinen and Dalrymple 1979). Therefore, it follows that the age we may expect for the base of our stratigraphic section should be 17.0–19.6 m.y. This is a wide and conservative age window (68% confidence level), but it is relatively insensitive to the various premises behind the sampling model. As broad as this estimate of age is, it nevertheless provides a useful temporal constraint on the correlation of our observed data with the time scale (fig. 5). Any correlation which places the base of the stratigraphic section outside of the 17.0–19.6 m.y. time range would be strongly suspect.

Finally, we pointed our previously that the Kamlial Formation consists of rocks that are mostly of reversed polarity. These can be identified with Chron 16, a rather long interval (15.1–17.6 m.y.) of mostly reversed polar-

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TABLE 1

ESTIMATE OF THE TIME INTERVAL (ΔT) REPRESENTED BY THE STRATIGRAPHIC INTERVAL BELOW N_0

Number of paleomagnetic sites (N)	Number of reversals found (R)	Probability $P = R/(N-1)$	Mean spacing of paleomagnetic sites[a] $\bar{S} = f(p)$	Expected time interval[b] $\Delta t = \bar{S}\bar{\tau}N$
92	27	.30	.17	8.3 ± 1.3 (1σ) (m.y.)

[a]From Johnson and McGee (1983, eq. 7 ($P = \bar{S}/(2\bar{S}+1)$). Assuming that samples are distributed with an exponential randomness over the stratigraphic interval, e.g., see figures 5 and 6, this paper.
[b]An estimate for $\bar{\tau}$ for the late Neogene is 1.2×10^5 years (Johnson and McGee 1983).

ity standing in marked contrast to the complex reversal sequence immediately following it in time (fig. 5). Our data do not show the short polarity zones within Chron 16 (figs. 4 and 5) because, with our relatively low sampling density (table 1), the probability that our sampling network would find such small events is quite low. For the same reason our sampling program was not sufficient to detect the fine structure of the Chron 12–13 interval or the Chron 15 interval (fig. 5). On the other hand, our sampling did uncover a small event between N_{12}–N_{13} (fig. 4) which is not contained in the time scale (fig. 5). Note, however, that this small event is based only on one class B paleomagnetic site, and therefore its validity may be questioned.

Because of the fission-track date in our section (9.5 ±0.5 m.y.), the distinctive signatures of Chrons 9 and 16, and the inferred time-span of our section (table 1), we are quite confident of our correlation with the magnetic reversal time scale (fig. 5). We note, however, that there may be other plausible correlations, and we cannot decisively preclude them at the present time. Nevertheless, when the intrinsic uncertainties in our sampling program (table 1) and the time scale (Lowrie and Kent 1983) are duly considered, the Chinji Village sequence contains a generally faithful record of the Earth's magnetic field.

SEDIMENTARY HISTORY

Our interpretation of the magnetic stratigraphy (fig. 5) indicates that fluvial sedimentation in the Chinji Village area began in the latter part of Chron 17, about 18.3 m.y. ago. This fluvial sedimentation continued at least until the upper boundary of Chron 8 (7.9 m.y.) at which point the sedimentary sequence is broken by an unconformity. In the Chinji Village area, therefore, fluvial sedimentation continued essentially uninterrupted for a minimum of 10.4 m.y., a significant span of geologic time. To gauge the magnitude of 10 m.y. consider that during such an interval the Indian Subcontinent would have drifted some 500 km northward closer to the Himalaya, carrying the Chinji Village depositional site with it. Thus, we may presume that the tectonic setting and/or sedimentary environment of the Chinji Village area changed substantially during the course of its Siwalik history. It is these changes we will look for and assess in the following pages.

From a stratigraphic perspective the data in figures 4 and 5 allow us to assign ages to the various formations in our section. As mapped the Kamlial Formation extends from the upper part of Chron 17 to the middle part of Chron 15, or nominally from 18.3 m.y. to 14.3 m.y. (according to the time scale of Mankinen and Dalrymple 1979). The Chinji Formation, the stratotype Chinji, extends from the middle of Chron 15 to about the lower third of Chron 10, or 14.3 m.y. to 10.8 m.y. (Mankinen and Dalrymple 1979). The stratotype for the Nagri Formation extends from the lower third of Chron 10 to about the Chron 8–9 boundary, or from 10.8 m.y. to 8.5 m.y. (Mankinen and Dalrymple 1979). These ages for the Nagri are consistent with those previously reported from five other areas in the Potwar Plateau (N. Johnson et al. 1982; Tauxe and Opdyke 1982). We use the term "consistent" advisedly because, as we previously reported (N. Johnson et al. 1982), formational boundary ages vary on the order of

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10^5 years from place to place over the Potwar area because of their time transgressive nature.

In figure 6 is a plot of sediment accumulation in the Chinji Village area versus time. As oriented the slope of the curve has the units of cm/time, or sedimentation rate. It is noteworthy that for the first 7 m.y. of Siwalik deposition, sediment accumulation in the Chinji area proceeded at a relatively uniform pace, averaging 0.12 mm/yr but perhaps with a slightly curvilinear aspect (fig. 7). Then at about 11 m.y. ago, sediment accumulation accelerated significantly, increasing to over 0.3 mm/yr by 8 m.y. ago.

The point in time when sediment accumulation becomes notably faster (~11 m.y.) coincides remarkably well with the Chinji-Nagri lithologic boundary (figs. 5 and 6). This increase in sedimentation rate at the Chinji-Nagri boundary has been previously described from other areas in the Potwar (Tauxe and Opdyke 1982; N. Johnson et al. 1982) so that it appears to be a pervasive phenomenon in this region.

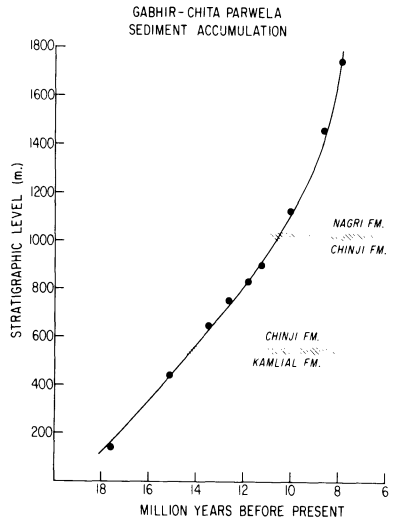


FIG. 6.—Sediment accumulation history of the Chita Parwela-Gabhir Kas stratigraphic section. Data points show boundaries between identified magnetic Chrons (figs. 4 and 5). Slope of the curve is sedimentation rate.

FLUVIAL CYCLES

Data on sedimentation rate presented in figure 6 have a bearing on the fluvial-hydrologic behavior of the study area. Fluvial cycles of fining-upward, channel/overbank deposits are well-developed in the Chinji Formation (Raza 1983), having a stratigraphic separation of 12–50 m. Given the 0.12 mm/yr mean sedimentation rate for Chinji time (fig. 6) this physical separation translates into a time separation of 1×10^5 yr to 4×10^5 yr. That is, during Chinji time the return period for the river channel at a given location was on the order of 10^5 years. During the deposition of the Middle and Upper Siwaliks the river return period was apparently faster, being on the order of 10^4 years (Opdyke et al. 1979; Barndt et al. 1978; Opdyke et al. 1982). It should be emphasized, however, that these faster return periods pertain to locations and time intervals other than those reported here. In any case, 10^4 or 10^5 years is a long time in the history and evolution of rivers. For example, 10^4–10^5 years would encompass a major continental glacial cycle, with the profound changes in hydrologic regimen that accompany such an event. This time scale (10^4–10^5 years) for a channel return period suggests that a major shift or migration of an entire drainage system is involved, not just a local variation within a steady-state meander belt.

Current direction studies in the Siwalik deposits of the Chinji Village area shed some light on this matter. Cross bedding studies in the Kamlial sands (Stix 1982) show that current directions were dispersed over an arc of 270° or more but generally indicate flow from the west to east. Similarly, cross-beds in the Chinji Formation (Sheik and Ibrahim Shah 1984) and the Nagri Formation (M. Asif, pers. comm.) show a consistent, sustained flow direction from west to east. Apparently for some 10 m.y., the river system which brought sediment into the Chinji Village area was fixed in its average spatial orientation. However, on a time scale of 10^4–10^5 years, the channel or channel complex of this river swept back and forth over the region. Each passage of the river channel over a given place marked another "fluvial cycle" in the sedimentary record.

In the Khaur area of the Potwar Plateau, Behrensmeyer and Tauxe (1982) have demonstrated that the principal river 8.1 m.y. ago ran from west to east, identical to the Chinji Village area. Raynolds (1980) also has shown that in the eastern Potwar during Upper

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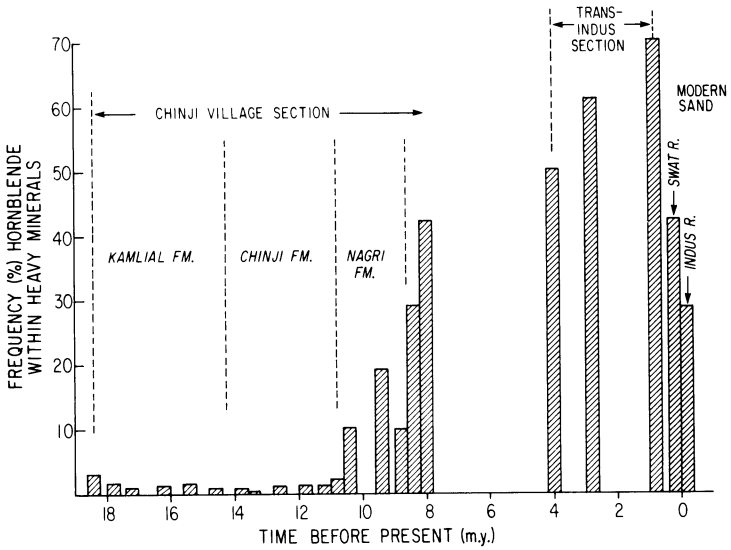


FIG. 7.—Abundance of blue-green hornblende in channel sand as a function of time. Ordinate is the proportion of blue-green hornblende within the heavy mineral suite, i.e., minerals with density greater than bromoform. Sandstone age assigned by position within the magnetic reversal time scale (figs. 4 and 5). Trans Indus sands are from Bain Pass section, described and dated by Khan (1983). Swat River sand obtained from river bed near Malakand, Indus River sand from near Attock.

Siwalik time the major river flowed from west to east. If these case studies are at all representative, it suggests that the fluvial system responsible for Siwalik sedimentation persisted in a west-to-east flow direction for some 16 m.y.

This west-to-east flow direction runs parallel to the present Himalayan belt, not away from it, an orientation which is repeated by the Ganges River of today. Pilgrim (1919) first hypothesized the existence of a "Siwalik River" much like the modern Ganges, i.e., a river running parallel to the mountain front, and ever since the Ganges system has served as a paradigm for Himalayan hydrology and sedimentation (Miall 1981). The current direction studies cited above support Pilgrim's (1919) concept of such a master river running parallel to the Himalaya; however, based on geomorphic evidence Pilgrim believed this river ran from east to west, contrary to what studies of the current direction now show. Nevertheless, the essential idea of using the Ganges system as an analog for Siwalik sedimentation seems justified.

Over much of the southern flank of the Himalaya today coalescing alluvial fans merge to form the Indo-Gangetic piedmont. The Kosi River is one of these quasi-distributary rivers; it leaves the mountain front near Mt. Everest and forms an alluvial fan (Das 1982). The Ganges parallels the Himalayan front with a series of Kosi-type tributaries on its north side. Both the Ganges and the Kosi-type rivers have braided habits and are subject to large lateral migrations. In particular, the Kosi River has migrated some 70 miles in the past 2 centuries (Das 1982). These data suggest to us that the return period for a Kosi-type river is probably 10^2–10^3 years, much too short for the fluvial cycles seen in Siwalik sediments. We believe, therefore, that a Ganges-type river with Ganges proportions was a more likely agent for the 10^4–10^5 year fluvial cycles which characterize Siwalik strata.

The studies of Behrensmeyer and Tauxe (1982) and Raynolds (1980) show the presence of a two-river system during Siwalik sedimentation. In both studies the principal river was oriented east-west, whereas a lesser river was oriented more north-south. Moreover, in both studies the major river deposited gray sands, and the lesser river deposited brown sands. By invoking the Pilgrim (1919) model we may then infer that the east-

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west river (gray sand) was of the Ganges-type, but the more north-south river (brown sand) was of the Kosi-type.

When considering the Siwalik sequence in its entirety, we may identify other fluvial cycles in addition to the 10^4–10^5 year cycle. The largest cycle is that of the Siwalik sequence itself. As reported here, the Siwalik sequence in the Chinji Village area is several kilometers thick and represents 10 m.y. or more of fluvial-tectonic history. An episode of fluvial sedimentation of this magnitude, caused by tectonism, is designated "first-order" by Schumm (1977, p. 330). In the case of the Siwalik sediments of the Potwar area, such first-order episodes span some 10^7 years.

A second category of fluvial event or cycle is exemplified by the fining-upward sequences of the Lower and Middle Siwaliks. These major fluvial cycles span on the order of 10^6 years and are superimposed on the basic first-order fluvial cycle. Schumm (1977, p. 330) defines such cycles as second-order. In the case of the Chinji Village area the Lower and Middle Siwalik units represent two consecutive fining-upward sedimentary cycles (second-order) which occurred as part of the basic Siwalik (first-order) episode of sedimentation. Second-order events such as these may reflect major, long-sustained shifts in local hydrologic regimen or changes in source area provenance.

Schumm (1977, p. 330) defines third-order fluvial cycles as those related to geomorphic thresholds, or step-functions. A definitive criteria for a third-order fluvial cycle is the development of channeling in fluvial deposits. The channel sand/overbank couplet that Allen (1965) defines as the basic fluvial cycle unit would thus be third-order. In the Siwalik system such third-order fluvial cycles have a 10^4–10^5 years periodicity.

Lastly, Schumm (1977, p. 330) cites storm and seasonal events as the cause of fourth-order fluvial cycles. In the overbank deposits of the Siwaliks system stratified clays, silts, and sands are frequently intermixed with paleosol horizons. These deposits are generally attributed to seasonal or flood events, which would qualify them as fourth-order fluvial cycles. In the Siwalik system such fourth-order cycles would have return periods spanning a wide range, probably 10^0–10^4 years depending on the space scale and the process involved.

SEDIMENTARY PETROLOGY AND TECTONICS

The source area for Siwalik sediments clearly has been the Himalayan highlands. The character and tectonics of these highlands in the past can be inferred in part from the sediments they shed. To this end we have collected 20 cross-bedded sands more or less uniformly throughout our Siwalik section. Thin sections of these sands reveal a marked increase in metamorphic minerals, especially blue-green hornblendes, in the upper part of the section. In addition, the grains in these younger sands appear to be more angular, and they contain fewer rock fragments than their older counterparts. These qualitative trends are confirmed by a quantitative analysis of the heavy mineral content of each sand (Gronseth 1983). Although several dozen minerals make up the heavy mineral ensemble, one mineral, blue-green hornblende, shows a conspicuous trend with time (fig. 7). For the first 7 m.y. it stays relatively constant at the 1–6% level but at the Chinji-Nagri boundary the relative proportion starts to increase rather abruptly. This point in time, 10.8 m.y., coincides with the increase in sedimentation rate previously noted (fig. 6).

Evidently, the Chinji-Nagri boundary signals an important change in sedimentary environment and provenance, change not limited to the Chinji Village area or the Potwar Plateau. The appearance of hornblende in Middle Siwalik sand has been noted in both the Trans-Indus region (Abid et al. 1983) and in northwest India (Raju and Dehadrai 1962). We have found also that the modern Indus River carries sand rich in blue-green hornblende, and further, that Pleistocene and Pliocene (Upper Siwalik) sand from the Trans-Indus region also is rich in blue-green hornblende (fig. 7). The clear implication is that a metamorphic terrane, specifically a hornblende-rich terrane, was unroofed over a wide front during Middle Siwalik time, and this terrane has contributed substantially to the heavy mineral content of Siwalik sand ever since. In the Chinji Village area this first-appearance datum is about 11 m.y. ago (fig. 7).

We may look to the Himalayan highlands for the cause of this compositional change. The present-day Swat River of northern Pakistan has sand whose heavy minerals are dominated by blue-green hornblende (fig. 7). The Swat River drains an old island arc

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(Kohistan Arc) terrane, where the principal rocks are ultramafic amphibolites and a granulitic norite. This island arc terrane, therefore, is the source for metamorphic minerals in modern river systems and also a plausible source for such minerals in the recent past. Dr. M. Qasim Jan has kindly pointed out to us that the granulite rocks of the island arc also contain unique, exceptionally pleochroic orthopyroxenes. An inspection of our samples shows that these distinctive orthopyroxenes are in fact present whenever the blue-green hornblendes are present. This co-occurrence further supports the concept of the Kohistan Arc as a past and present source terrane for blue-green hornblendes.

Lastly, isotopic data from the Pakistani Himalaya also shed some light on this problem. Zeitler (1983, p. 160) has shown that an episode of rapid and sustained uplift, involving the central and northeast Kohistan Arc, began about 15 m.y. ago. The focus of this uplift activity has been the Nanga Parbat-Hunza region, where over 10 km of uplift has occurred in less than 10 m.y. (Zeitler et al. 1982). Uplift of this magnitude and rapidity has had a profound effect on the extant drainage system, the present Indus River, and its immediate predecessor. The rocks of the Kohistan Arc have been stripped back and away from these uplifted areas, leaving only erosional remnants behind as outliers (Malinconico 1982). The present Indus Gorge through the Nanga Parbat massif (5 km of proximal relief) is undoubtedly a result of this recent uplift activity (Zeitler et al. 1982).

It seems then that the gray sand of the contemporary Indus River is readily explained by the bedrock composition and uplift history of its headwater reaches. That these conditions prevailed for the past 11 m.y. is plausible, and in fact supported by the Siwalik sedimentary record.

Several important conclusions about the tectonic history of the Chinji Village area can be drawn from the above data. Of first importance is the evidence that molassic sedimentation did not actually reach the study area until 18.3 m.y. ago, even though the Himalayan orogeny began much earlier (~40 m.y.). Prior to 18.3 m.y. the Chinji Village site was a subdued, karst topography, not affected by the continental collision taking place further to the north. By 18.3 m.y., however, either the Chinji Village area had moved north, or the edge of molassic deposition had moved south far enough so that fluvial aggradation could begin. Our data showing a systematic increase in sedimentation rate over time (fig. 6) suggests that the Chinji Village area gradually moved closer to its sediment source over time, the overall result of the northward drift of India.

It is relevant to consider in this context just where molassic sediments that are older and younger than those at Chinji Village are located. Molassic sediments older than 18.3 m.y. are limited in extent and are found generally to the north of the Chinji Village area. These older rocks are also heavily folded, uplifted, and eroded. Significantly, the Chinji Village area today is no longer a site of deposition, but rather a site of erosion and a source of reworked, recycled sediment. Some 2 m.y. ago the Chinji Village area began its first involvement in the Himalayan collision process. The form of this involvement was the Salt Range Thrust, which effectively stopped any further sediment accumulation and started the present erosional activity. Raynolds (1980) has shown that the brown sands deposited during Upper Siwalik time in the Potwar area are reworked, older Siwalik deposits.

With regard to younger and modern molassic sediments the principal depositional centers lie far to the south of the Salt Range and the Chinji Village area. In other words, the sedimentary record shows a spatially and temporally distributed progression of molassic depositional centers. Over time these depositional centers have migrated gradually northward; or conversely, the edge of molassic sedimentation has systematically migrated southward. If the current model of Himalayan tectonism is correct and the northward drift of India continues at the present pace (McKenzie and Sclater 1971), we may anticipate that the Chinji Village area will be incorporated and consumed by the Himalayan collision zone within a few million years. When this happens, the present Siwalik sediments of Chinji Village will have prospects of being cannibalized and recycled into future Siwalik brown sands. This process has in fact already started, as witnessed by the eroded badlands of the Chinji Village area.

In the long-range view then, the Siwalik sequence in the Chinji Village area represents

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just one ephemeral stage in a dynamic system of landforms, sediments, and tectonics. In the course of its northward drift the Indian Plate has acted like a conveyor belt, bringing a continuous series of depositional sites, including the Chinji Village area, along with it. During its northward ride, the Chinji Village site has been converted slowly from a karst terrane into a depositional terrane, and most recently into a thrust belt and source terrane. Our chronologic data from Chinji Village suggest that the life cycle for each depositional site spans some 20 m.y.

CONCLUSIONS

Paleomagnetic data from the Chinji Village area have chronological, fluvial, and tectonic implications:

1. The local magnetic polarity stratigraphy is securely correlated with the magnetic reversal scale by means of isotopic data, distinctive magnetic signatures, and statistical inferences.

2. Siwalik sedimentation in the Chinji Village area spans at least 10.4 m.y., starting at 18.3 m.y. and continuing until at least 7.9 m.y.

3. In the study area deposition of the Kamlial Formation took place from 18.3 m.y. to 14.3 m.y.; the stratotype Chinji Formation from 14.3 m.y. to 10.8 m.y.; the stratotype Nagri Formation from 10.8 m.y. to 8.5 m.y.

4. Siwalik strata are a composite of four distinct orders of fluvial cycles (Schumm, 1977), each order having different causes and different periodicities: (i) First-order. The Siwalik sequence at any given location; time scale on the order of 10^7 years; causal agent is tectonism. (ii) Second-order. Massive fining-upward sequences with a time scale of 10^6 years; in the Chinji Village area represented by the Lower Siwalik Subgroup (Kamlial and Chinji formations) and the Middle Siwalik Subgroup (Nagri and Dhok Pathan formations); causal agent is tectonism and/or climate. (iii) Third-order. Basic fining-upward unit defined by Allen (1965); caused by river migration and channeling; in the Siwalik sequence having a periodicity of 10^4–10^5 years. (iv) Fourth-order. Caused by seasonal or storm events or soil-forming processes; wide-ranging periodicity, 10^0–10^4 years in the Siwalik sequence.

5. Over a 10.4 m.y. time span Siwalik deposition in the Chinji Village area was associated with a river system which flowed essentially from west to east. This east-west orientation coincides rather well with Pilgrim's (1919) model of the "Siwalik River" system. This model is basically an analogy with the modern Ganges River system.

6. Siwalik sedimentation rates in the Chinji Village area increase systematically over time from 0.12 to 0.30 mm/yr. A pronounced increase in sedimentation rate occurs at about 11 m.y.; simultaneously blue-green hornblendes also appear in Siwalik channel sands. These events are attributed to the initial uplift of the Nanga Parbat region and the associated erosion of ultramafic rocks in northern Pakistan (Zeitler 1983).

7. In a geologic sense Siwalik strata in the Chinji Village area are not likely to survive long as a sedimentary record. Given the present rate of northwind drift of the Indian Plate, the Chinji Village site will arrive at the main Himalayan collision zone within a few million years, where it will be consumed by the collision process. So, although the Siwalik sequence in the Chinji Village area has survived essentially intact for some 18 m.y., it is nearing the end of its life-cycle as a sedimentary unit.

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